CEUE203 - OOP25AIML034

PRACTICAL – 6

Aim:

To develop Java programs using interfaces, functional interfaces, default methods, lambda expressions, anonymous classes, and marker interfaces to control devices, broadcast notifications, and apply discount rules to a list of prices.

Code Screenshot:

```
interface Switchable {
    void on();
    void off();

    default void toggle() {
        on();
    }
}

class Fan implements Switchable {
    public void on() {
        System.out.println("Fan is ON");
    }

    public void off() {
```

```
        System.out.println("Fan is OFF");
    }
}

class Light implements Switchable {
    public void on() {
        System.out.println("Light is ON");
    }

    public void off() {
        System.out.println("Light is OFF");
    }
}

interface SwitchDecision {
    boolean maySwitchOn(String device, int hour);
}

public class Main {
    public static void main(String[] args) {
        Switchable[] devices = {
            new Fan(), new Light()
```

```
};

System.out.println("Toggle devices:");
for (Switchable device : devices) {
    device.toggle();
}

SwitchDecision anonymous = new SwitchDecision() {
    public boolean maySwitchOn(String device, int hour) {
        return hour >= 6 && hour < 22;
    }
};

SwitchDecision lambda = (device, hour) ->
        hour >= 6 && hour < 22;

System.out.println("\nAnonymous class:");
System.out.println("Fan at 10:00: " +
        anonymous.maySwitchOn("Fan", 10));

System.out.println("\nLambda expression:");
System.out.println("Light at 23:00: " +
```

```
            lambda.maySwitchOn("Light", 23));
    }
}
```

2.

```
@FunctionalInterface
interface Notifier {
    void send(String message);
}

interface Urgent {
}

class EmailSender implements Notifier, Urgent {
    public void send(String message) {
        System.out.println("Email: " + message);
    }
}

class SmsSender implements Notifier {
    public void send(String message) {
        System.out.println("SMS: " + message);
```

```
    }
}


public class Main {
    public static void main(String[] args) {
        Notifier email = new EmailSender();
        Notifier sms = new SmsSender();

        Notifier[] senders = {
            email,
            sms,
            (message) ->
                System.out.println("App notification: " + message)
        };

        String message = "Your order has been shipped.";

        System.out.println("Normal broadcast:");
        for (Notifier sender : senders) {
            sender.send(message);
        }
```

```
        System.out.println("\nUrgent broadcast:");
        for (Notifier sender : senders) {
            if (sender instanceof Urgent) {
                sender.send(message);
                sender.send(message);
            }
        }
    }
}
```

3.

```
import java.util.*;

interface DiscountRule {
    double apply(double price);
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        List<Double> prices = Arrays.asList(
```

```
    1000.0, 2000.0, 500.0
);

DiscountRule tenPercent = price -> price * 0.90;
DiscountRule twentyPercent = price -> price * 0.80;
DiscountRule noDiscount = price -> price;

System.out.println("Choose discount:");
System.out.println("1. 10% discount");
System.out.println("2. 20% discount");
System.out.println("3. No discount");
System.out.print("Enter choice: ");

int choice = sc.nextInt();

DiscountRule selected;

switch (choice) {
    case 1:
        selected = tenPercent;
        break;
    case 2:
```

```
        selected = twentyPercent;
        break;
    case 3:
        selected = noDiscount;
        break;
    default:
        System.out.println("Invalid choice.");
        sc.close();
        return;
}

double total = 0;

System.out.println("\nDiscounted prices:");
for (double price : prices) {
    double finalPrice = selected.apply(price);

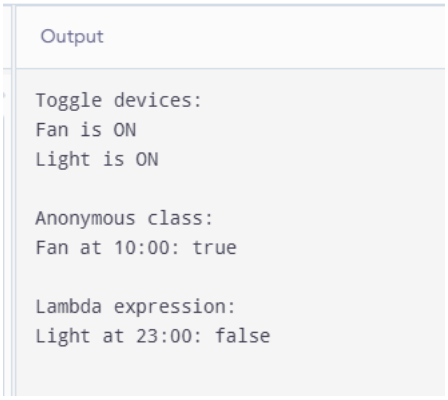
    System.out.printf(
        "Original: Rs. %.2f, Final: Rs. %.2f%n",
        price, finalPrice
    );
```

```
            total += finalPrice;
        }

        System.out.printf("Total: Rs. %.2f%n", total);

        sc.close();
    }
}
```

Output Screenshot:

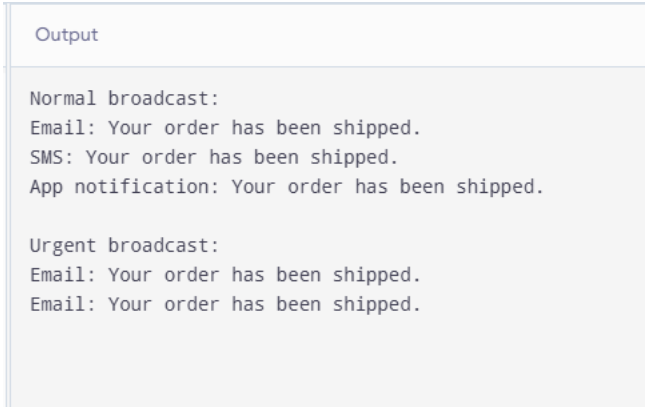


```
Output

Toggle devices:
Fan is ON
Light is ON

Anonymous class:
Fan at 10:00: true

Lambda expression:
Light at 23:00: false
```



```
Output

Normal broadcast:
Email: Your order has been shipped.
SMS: Your order has been shipped.
App notification: Your order has been shipped.

Urgent broadcast:
Email: Your order has been shipped.
Email: Your order has been shipped.
```

Output

```
Choose discount:
1. 10% discount
2. 20% discount
3. No discount
Enter choice: 1

Discounted prices:
Original: Rs. 1000.00, Final: Rs. 900.00
Original: Rs. 2000.00, Final: Rs. 1800.00
Original: Rs. 500.00, Final: Rs. 450.00
Total: Rs. 3150.00
```